

Term 3 Robotics Challenge 2026 (Control) [20pts]

Trial Runs #2 Checklist: June 2, 2026

Grading Policy & Rules

- Grading is based on the full completion of the 8 control and navigation tasks detailed below.
- Each team is allocated exactly 20 minutes in total (4 minutes for setup/removal and 16 minutes of active testing within their task slot).
- A maximum of 2 trials (1 reset) is allowed per task.
- Penalty: A 1-point deduction will be strictly applied for each manual intervention.

Checklists

1. Standard Line Tracking [2pts]

Task: The robot must navigate from “Start” (e.g., its deployment area) to the “Destination” (e.g., exit door), following a solid black line smoothly without losing tracking.

- **Success (2 pts):** Successfully navigates the full path within 30 seconds without permanently losing the line.
- **Partial Credit (1 pt):** Completes at least half of the tracking line before a failure occurs.
- **Failure (0 pts):** Stops prematurely, departs entirely from the marked track, or fails to advance.

2. Intersection & Tag Alignment [2pts]

Task: The robot must follow a branched line inside the base, correctly identify the bifurcation, and deliberately stop its line-following behaviour over the designated 'B' RFID tag to request exit clearance.

- **Success (2 pts):** Autonomously follows the correct line branch and centers its sensor array over the RFID tag within 45 seconds.
- **Partial Credit (1 pt):** Follows the track but misses the branch/tag alignment, requiring manual reset.
- **Failure (0 pts):** Ignores the path intersection entirely, drives off-course, or fails to halt over the target.

3. Solid Grid Navigation (Left Half Arena) [3pts]

Task: The robot must operate on the left side of the Main Arena, using its sensor array to follow solid grid lines that connect the underlying RFID nodes.

- **Success (3 pts):** Smoothly executes grid-line tracking across multiple nodes for 1 meter within 45 seconds.
- **Partial Credit (2 pts):** Covers half the distance or experiences minor tracking oscillations but recovers autonomously.
- **Partial Credit (1 pt):** Covers one third of the distance or experiences significant tracking oscillations but manages to recover.
- **Failure (0 pts):** Loses tracking on the solid grid lines immediately.

4. Open-Field Dead Reckoning (Right Half Arena) [3pts]

Task: The robot must navigate to the right half of the arena, where no solid floor markings exist, moving reliably between spaced RFID hole nodes using dead reckoning, odometry, or localised distance checking.

- **Success (3 pts):** Accurately traverses from one node hole to another across the unlined surface field within 45 seconds.
- **Partial Credit (2 pts):** Minor position drifts but successfully encounters a nearby node hole.
- **Partial Credit (1 pts):** Significant position drifts but successfully encounters a nearby node hole.
- **Failure (0 pts):** Drifts completely off course, missing the node hole entirely.

5. Ramped Incline/Decline Control [2pts]

Task: The robot must successfully ascend or descend the airlock access tunnels, maintaining stable velocity and balance control on the slope.

- **Success (2 pts):** Traverses the entirety of the ramp smoothly without stalling, sliding backward, or tipping over within 45 seconds.
- **Partial Credit (1 pt):** Ascends or descends with severe speed variations or minor slipping but safely completes the grade transition.
- **Failure (0 pts):** Stalls due to motor limits, tips backward, or slips uncontrollably down the incline.

6. Wall Following [2pts]

Task: Utilising its distance sensors, the robot must smoothly track along the solid walls of the arena perimeter or airlock boundaries at a fixed offset distance without experiencing physical collisions.

- **Success (2 pts):** Maintains a consistent, stable distance offset over a 1-meter tracking path within 45 seconds.
- **Partial Credit (1 pt):** Exhibits minor drifting or bumps into the wall surface slightly but manages to self-correct and continue.
- **Failure (0 pts):** Hard crashes repeatedly into the wall boundary or drifts completely away from the sensor tracking threshold.

7. Obstacle Detection and Avoidance [3pts]

Task: The robot must navigate through obstacles placed between node holes without colliding with them.

- **Success (3 pts):** Detects and calculates a path around the obstacle safely within 60 seconds.
- **Partial Credit (2 pt):** Suffers minor, transient contact with the obstacles but preserves its trajectory vector to squeeze past the hazard line.
- **Partial Credit (1 pt):** Experiences severe structural scraping or continuous collision against the obstacles but maintains enough forward drive to pass the hazard line.
- **Failure (0 pts):** Directly collides with the obstacle, requiring manual reset.

8. Touch-Based Robot Revival [3pts]

Task: Given the information of a stationary stranded robot, the active robot must execute a controlled deceleration approach and make stable mechanical contact with the target to rescue it.

- **Success (3 pts):** Executes a smooth deceleration profile and successful contact the target within 60 seconds.
- **Partial Credit (2 pt):** Successfully triggers the mechanical revival state change but exhibits contact oscillations or localised chatter.
- **Partial Credit (2 pt):** Successfully triggers the mechanical revival state change but exhibits a high-impact collision overshoot or contact oscillations.
- **Failure (0 pts):** The robot completely past or glances off the side of the stranded robot.

Performance Log Sheet

Task	Attempt 1	Attempt 2	Interventions (-1pt)	Total Score
1. Standard Line Tracking				/ 2 pts
2. Intersection & Tag Alignment				/ 2 pts
3. Solid Grid Navigation				/ 3 pts
4. Open-Field Dead Reckoning				/ 3 pts
5. Ramped Incline/Decline Control				/ 2 pts
6. Wall Following				/ 2 pts
7. Obstacle Detection and Avoidance				/ 3 pts
8. Touch-Based Robot Revival				/ 3 pts
Total Marks Earned				/ 20 pts

Marker Sign-off: _____

Date: June 2, 2026